Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An EM wave in vacuum has frequency 5×10^{9} Hz. Use $c=3.00\times10^{8}$ m/s. Find wavelength and name the spectral region.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: range = c Δt/2; c = 3.0×10^{8} m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range=c Δt/2=300 m. The measured time includes the outward and return paths.

Your turn: Draw the round trip and label the conversion 2 microseconds=2×10^{-6} s.



Marzano check - circle: ① ② ③

03 Read and interpret the model

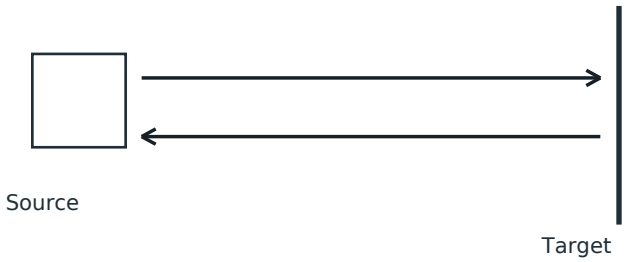
Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?



Marzano check - circle: ① ② ③

04 Apply the physics

A stationary radar receives an echo after 0.040 s. Use $c = 3\times10^{8}$ m/s to find range. Separately, its wavelength is 0.10 m and Doppler-shift magnitude is 400 Hz. Use $|v_r| = |\Delta f|\lambda/2$ to find radial speed. Label which measurement supports each result.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? Sketch another possible motion.

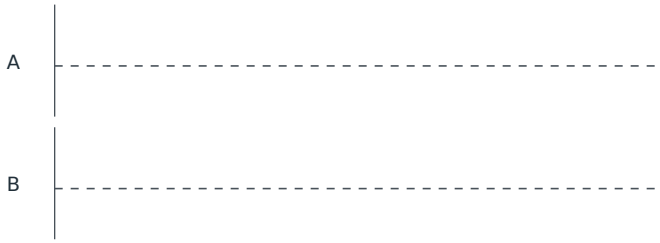
Add a velocity perpendicular to the line of sight.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave travels 12 m in 0.4 s at frequency 60 Hz. Find speed and wavelength.
Label a wave sketch to support the calculation or comparison.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

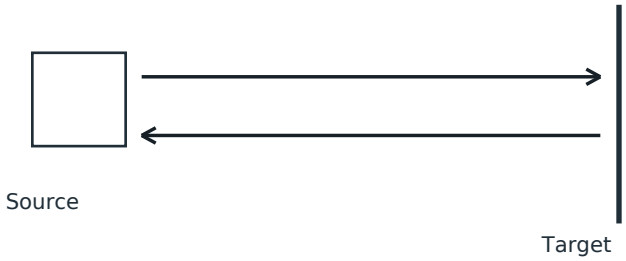
Two targets at the same range have different Doppler shifts. Explain why they can have same time-delay but different motions.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Radar range changes from 1500 to 1440 m over 0.5 s. Find average radial velocity taking outward positive. Explain why total speed may be larger.



Marzano check - circle: 1 2 3